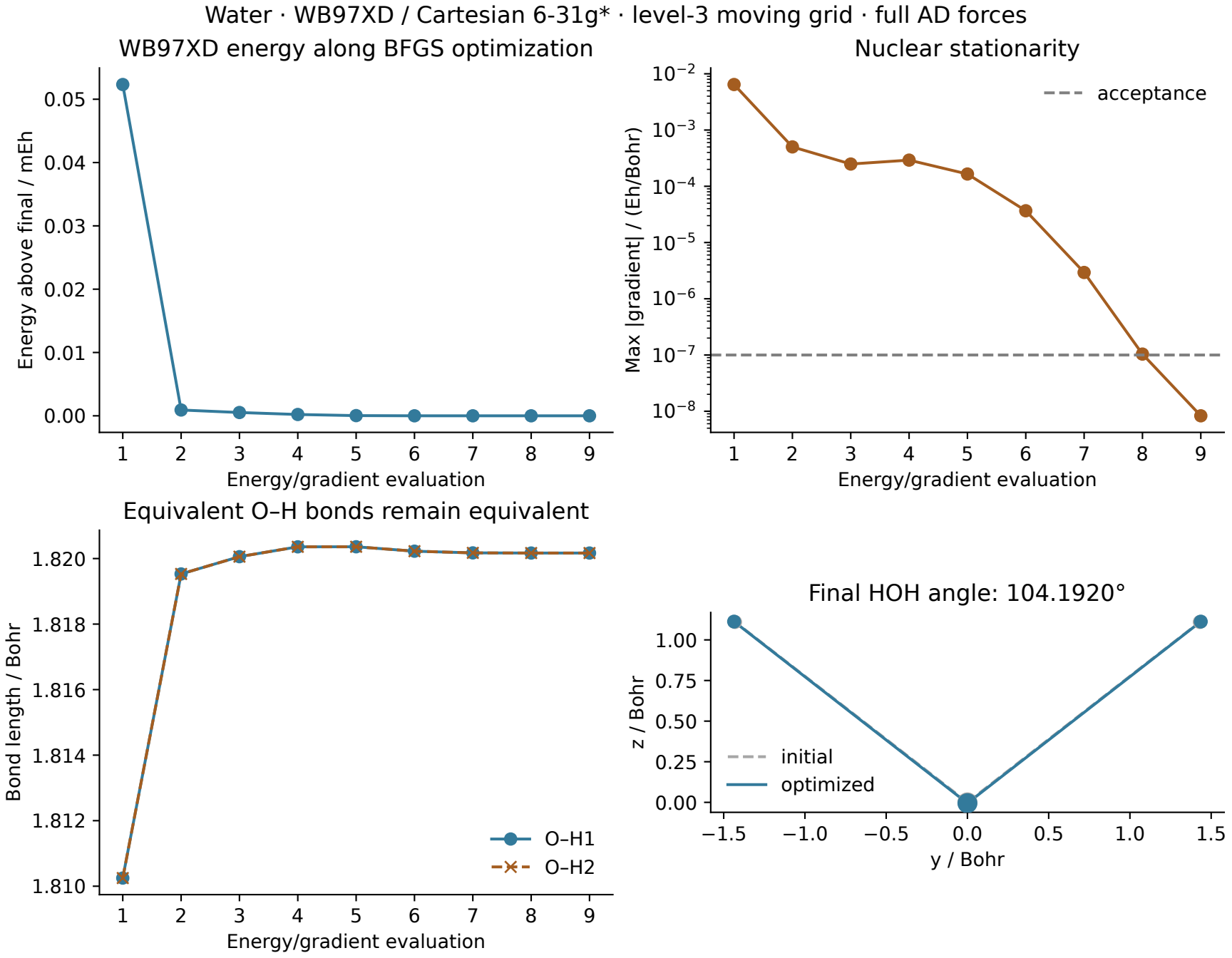
Water · WB97XD / Cartesian 6-31g* · level-3 moving grid · full AD forces
WB97XD energy along BFGS optimization
Energy above final / mEh
0.05
0.04
0.03
0.02
0.01
0.00
1
2
3
4
5
6
7
8
9
Energy/gradient evaluation
Nuclear stationarity
Max |gradient| / (Eh/Bohr)
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
acceptance
Energy/gradient evaluation
Equivalent O–H bonds remain equivalent
Bond length / Bohr
1.820
1.818
1.816
1.814
1.812
1.810
O–H1
O–H2
Energy/gradient evaluation
Final HOH angle: 104.1920°
z / Bohr
1.00
0.75
0.50
0.25
0.00
initial
optimized
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
y / Bohr